

Divide and Conquer – Searching & Basic Problems

Binary Search, Finding Maximum and Minimum using Divide and Conquer

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Question 1: Search an Element in a Sorted Array

Given a sorted array of integers and a key value, use the Binary Search (Divide and Conquer) method to determine whether the key is present in the array. If found, print its position; otherwise, print "Element not found".

Test Case 1

Input

5
10 20 30 40 50
30

Expected Output

Element found at position 3

Test Case 2

Input

5
10 20 30 40 50
25

Expected Output

Element not found

Question 2: Find the Index of a Given Number

Given a sorted array and a target number, find the index of the target using Binary Search. Assume array indexing starts from 0.

Test Case 1

Input

6
2 4 6 8 10 12
8

Expected Output

Index = 3

Test Case 2

Input

6
2 4 6 8 10 12
5

Expected Output

Index = -1

Question 3: Count the Number of Iterations in Binary Search

Given a sorted array and a key, perform Binary Search and display the number of iterations required to find the element.

Test Case 1

Input

7
5 10 15 20 25 30 35
25

Expected Output

Element found
Iterations = 2

Test Case 2

Input

7
5 10 15 20 25 30 35
40

Expected Output

Element not found
Iterations = 3

Question 4: Find the First Occurrence of an Element

Given a sorted array containing duplicate elements, find the first occurrence of a given key using Binary Search.

Test Case 1

Input

8

1 2 2 2 3 4 5 6

2

Expected Output

First occurrence at index 1

Test Case 2

Input

8

1 2 2 2 3 4 5 6

7

Expected Output

Element not found

Question 5: Find the Last Occurrence of an Element

Given a sorted array containing duplicate elements, find the last occurrence of a given key using Binary Search.

Test Case 1

Input

8

1 2 2 2 3 4 5 6

2

Expected Output

Last occurrence at index 3

Test Case 2

Input

8

1 2 2 2 3 4 5 6

9

Expected Output

Element not found

Question 6: Highest and Lowest Temperature Recorded

A weather station records temperatures over a week. Find the **minimum** and **maximum** temperature using the Divide and Conquer approach.

Input

7

32 28 35 25 31 29 37

Expected Output

Minimum Temperature = 25

Maximum Temperature = 37

Question 6: Best and Worst Student Marks

A teacher wants to identify the highest and lowest marks scored in a class using Divide and Conquer.

Input

8

78 92 65 88 95 72 81 69

Expected Output

Minimum Mark = 65

Maximum Mark = 95

Question 8: Stock Price Analysis

An investor wants to know the minimum and maximum stock prices recorded during a trading week.

Input

6

450 520 480 610 430 590

Expected Output

Minimum Stock Price = 430

Maximum Stock Price = 610

Question 9: Fastest and Slowest Race Times

A sports analyst needs to find the fastest (minimum) and slowest (maximum) race completion times.

Input

9

14 12 18 11 15 13 17 16 10

Expected Output

Fastest Time = 10

Slowest Time = 18

Question 10: Largest and Smallest Building Heights**Input**

10

120 150 98 175 140 165 110 190 130 145

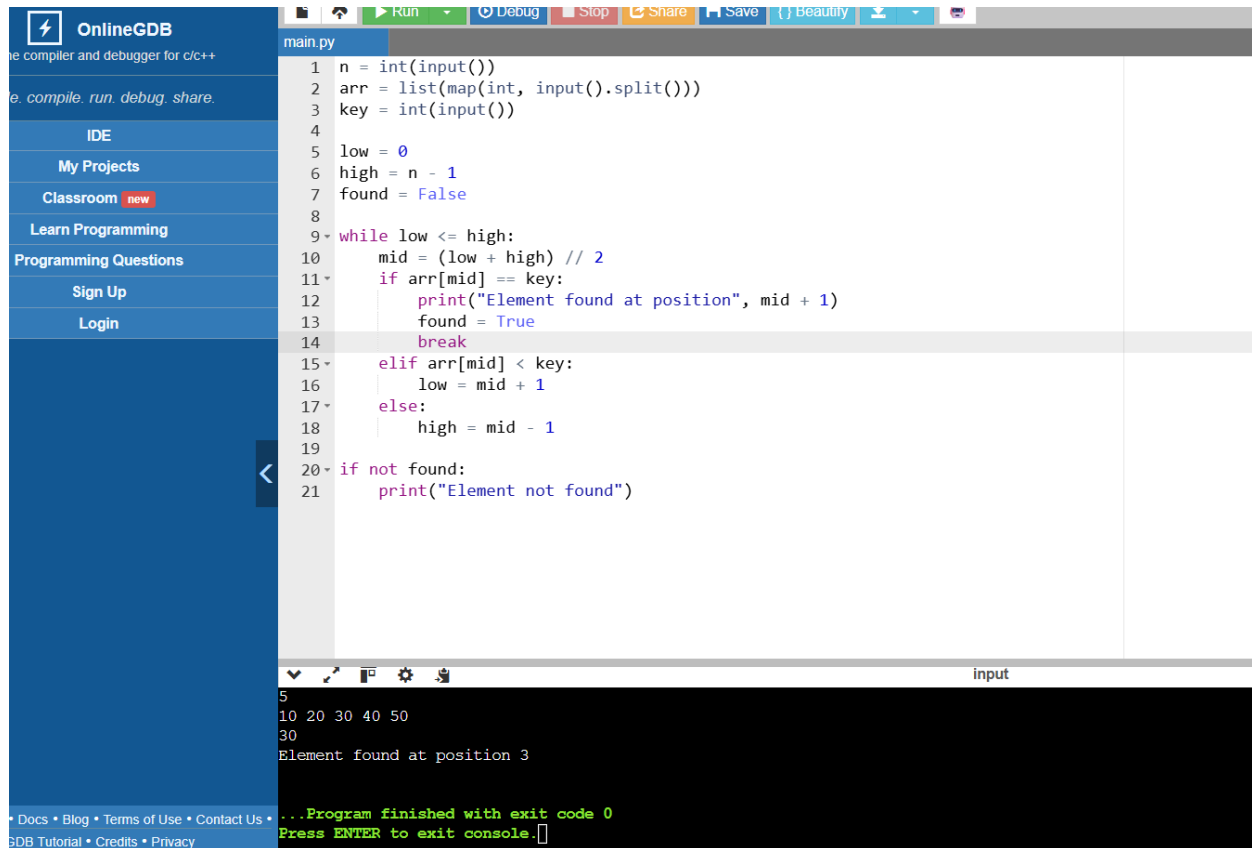
Expected Output

Minimum Height = 98

Maximum Height = 190

OUTPUTS:

Question 1: Search an Element in a Sorted Array



The screenshot shows the OnlineGDB IDE interface. On the left is a sidebar with navigation links: IDE, My Projects, Classroom (marked 'new'), Learn Programming, Programming Questions, Sign Up, and Login. The main editor area displays a Python file named 'main.py' containing a binary search algorithm. The code takes an integer 'n' and a list of integers 'arr' as input, then searches for a 'key' using a while loop. The output console at the bottom shows the input '10 20 30 40 50' and '30', followed by the output 'Element found at position 3'. The console also shows the program finished with exit code 0 and a prompt to press ENTER to exit.

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 found = False
8
9 while low <= high:
10     mid = (low + high) // 2
11     if arr[mid] == key:
12         print("Element found at position", mid + 1)
13         found = True
14         break
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 if not found:
21     print("Element not found")
```

input

5
10 20 30 40 50
30
Element found at position 3

...Program finished with exit code 0
Press ENTER to exit console.

Question 2: Find the Index of a Given Number

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main.py

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 target = int(input())
4
5 low = 0
6 high = n - 1
7 index = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11     if arr[mid] == target:
12         index = mid
13         break
14     elif arr[mid] < target:
15         low = mid + 1
16     else:
17         high = mid - 1
18
19 print("Index =", index)
```



input

```
6
2 4 6 8 10 12
8
Index = 3

...Program finished with exit code 0
Press ENTER to exit console.
```

Question 3: Count the Number of Iterations in Binary Search

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main.py

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 iterations = 0
8 found = False
9
10 while low <= high:
11     iterations += 1
12     mid = (low + high) // 2
13
14     if arr[mid] == key:
15         found = True
16         break
17     elif arr[mid] < key:
18         low = mid + 1
19     else:
20         high = mid - 1
21
22 if found:
23     print("Element found Iterations =", iterations)
24 else:
25     print("Element not found Iterations =", iterations)
```



input

```
7
5 10 15 20 25 30 35
40
Element not found Iterations = 3

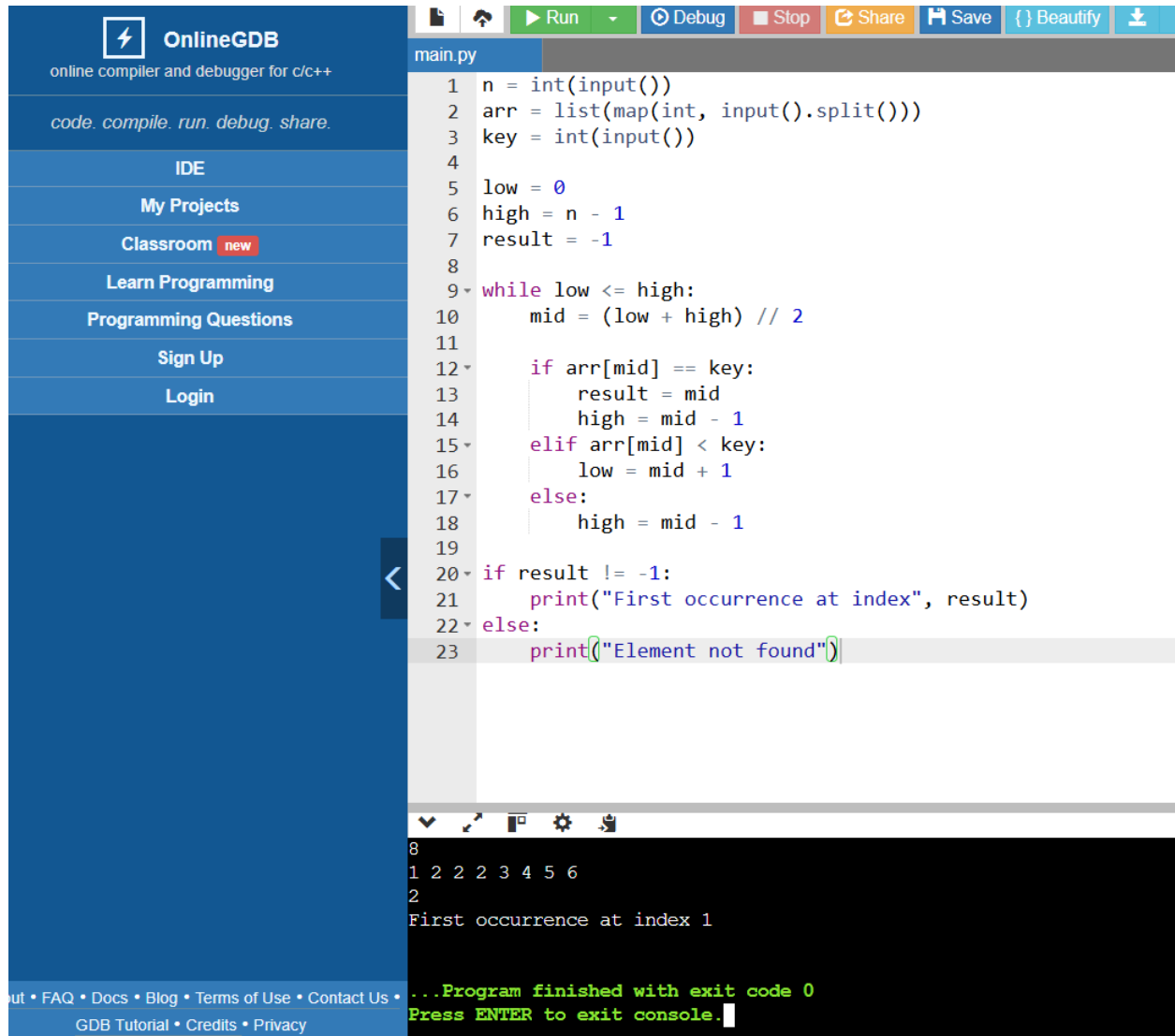
...Program finished with exit code 0
Press ENTER to exit console.
```

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Question 4: Find the First Occurrence of an Element



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```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 result = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         result = mid
14         high = mid - 1
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 if result != -1:
21     print("First occurrence at index", result)
22 else:
23     print("Element not found")
```

Below the code editor is a console window showing the program's execution. It displays the input '1 2 2 2 3 4 5 6' and the output 'First occurrence at index 1'. At the bottom, a message states '...Program finished with exit code 0' and 'Press ENTER to exit console.'

Question 5: Find the Last Occurrence of an Element

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main.py

```
1 n = int(input())
2 arr = list(map(int, input().split()))
3 key = int(input())
4
5 low = 0
6 high = n - 1
7 result = -1
8
9 while low <= high:
10     mid = (low + high) // 2
11
12     if arr[mid] == key:
13         result = mid
14         low = mid + 1
15     elif arr[mid] < key:
16         low = mid + 1
17     else:
18         high = mid - 1
19
20 if result != -1:
21     print("Last occurrence at index", result)
22 else:
23     print("Element not found")
```



```
8
1 2 2 2 3 4 5 6
2
Last occurrence at index 3

...Program finished with exit code 0
Press ENTER to exit console.
```

Question 6: Highest and Lowest Temperature Recorded

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main.py

```
1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Minimum Temperature =", minimum)
24 print("Maximum Temperature =", maximum)
```

7

32 28 35 25 31 29 37

Minimum Temperature = 25

Maximum Temperature = 37

...Program finished with exit code 0

Press ENTER to exit console.

Run

Debug

Stop

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Save

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Question 7: Best and Worst Student Marks

The screenshot shows the OnlineGDB IDE interface. On the left is a sidebar with navigation links: IDE, My Projects, Classroom (marked 'new'), Learn Programming, Programming Questions, Sign Up, and Login. The top of the IDE has a toolbar with icons for Run, Debug, Stop, Share, Save, and Beautify. The main editor area displays a Python file named 'main.py' with the following code:

```
1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Minimum Mark =", minimum)
24 print("Maximum Mark =", maximum)
```

Below the code editor is a terminal window showing the program's execution. It displays the input array '78 92 65 88 95 72 81 69', the calculated 'Minimum Mark = 65', and the 'Maximum Mark = 95'. The terminal also shows the message '...Program finished with exit code 0' and 'Press ENTER to exit console.'.

Question 8: Stock Price Analysis


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main.py

```

1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Fastest Time =", minimum)
24 print("Slowest Time =", maximum)

```






```

9
14 12 18 11 15 13 17 16 10
Fastest Time = 10
Slowest Time = 18

...Program finished with exit code 0
Press ENTER to exit console.

```

Question 10: Largest and Smallest Building Heights

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main.py

```
1 def min_max(arr, low, high):
2     if low == high:
3         return arr[low], arr[low]
4
5     if high == low + 1:
6         if arr[low] < arr[high]:
7             return arr[low], arr[high]
8         else:
9             return arr[high], arr[low]
10
11     mid = (low + high) // 2
12
13     min1, max1 = min_max(arr, low, mid)
14     min2, max2 = min_max(arr, mid + 1, high)
15
16     return min(min1, min2), max(max1, max2)
17
18 n = int(input())
19 arr = list(map(int, input().split()))
20
21 minimum, maximum = min_max(arr, 0, n - 1)
22
23 print("Minimum Height =", minimum)
24 print("Maximum Height =", maximum)
```

inp

6
450 520 480 610 430 590
Minimum Height = 430
Maximum Height = 610

...Program finished with exit code 0
Press ENTER to exit console.

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